Q1.

Find any relative extrema of
$f(x) = x\sqrt{9 - x^2}$

State where the function is increasing or decreasing and determine points of inflection.

Q2.

Find the absolute maximum and minimum values of
$f(x) = \frac{x^2}{x^3+2}$

on $[0, +\infty)$, if they exist.

Q3.

The total cost and total revenue functions are
$C(x) = x^3 - 12x^2 + 50x + 200$
$R(x) = x(120 - 3x)$

a) Write the profit function $P(x)$ and determine the level of output x for which profit is maximized.

b) Find the marginal revenue $MR(x)$ and marginal cost $MC(x)$. Verify that $MR = MC$ at maximum profit.

Q4.

A rectangular box with a square base and a cover is to have a volume of $2000\ \text{cm}^3$.

If the cost per cm^2 is:

- bottom: \$4
- top: \$2
- sides: \$3

Find the dimensions that minimize the cost.